

HI-VQE on Li_2S at 24 Qubits

A variationally bounded method used as a
certificate on its own reference

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Abstract

The Handover Iterative Variational Quantum Eigensolver (HI-VQE, Pellow-Jarman *et al.*, arXiv:2503.06292) is implemented on Qiskit and applied to that paper’s own benchmark system: Li_2S in CAS(12e,12o)/STO-3G, 24 qubits, 853,776 determinants, along the Li–S dissociation coordinate. Thirteen geometries were computed, together with a classical selected-CI control, an ablation of the algorithm’s two halves, and MP2/CCSD/CCSD(T) baselines in the identical active space.

Eleven of thirteen geometries land inside chemical accuracy of exact diagonalisation, at a mean error of 0.081 mHa over those points, using a mean of 2.25% of the determinant space. Hartree–Fock misses the same curve by up to 168 mHa. The dissociation energy computed from the curve is 103.89 kcal/mol against exact diagonalisation’s 103.94 — an error of 0.051 kcal/mol — and because a dissociation energy is a difference, that is a stricter consistency test than any single total energy. The sampling circuit requires one measurement setting where a conventional VQE on this Hamiltonian would require 15,697 Pauli words per optimiser step.

The two geometries that fail do so in a way worth reporting. At 3.20 and 3.60 Å the HI-VQE energy lies *below* the CASCI reference, by 8.50 and 24.24 mHa. For a variational method whose energy is the lowest eigenvalue of PHP for a projector P , that is impossible against a correct reference — so the finding is not that HI-VQE erred, but that the reference at those two points is not the ground state of the sector it was intended to diagonalise. Two independent arguments confirm this: the variational bound itself, and the non-monotonicity it induces in the reference potential-energy curve. Both failing points return $\langle \hat{S}^2 \rangle = 2.00$, the triplet.

The ablation is reported with its negative reading intact: a classical selected-CI control with no circuit at all reaches 0.141 mHa at the equilibrium geometry, against the quantum-sampled 0.076 mHa. At this size the classical expansion is sufficient, and the table exists so that the statement is checkable rather than assumed.

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1 Introduction

1.1 What HI-VQE changes

A conventional VQE asks a quantum device for an *energy*. That requires measuring every Pauli word in the Hamiltonian, every iteration. For this system the published work counts 15,697 Pauli words — 15,697 measurement settings per optimiser step — and `run.py paulis` recomputes that figure from the cached integrals, so it is checkable rather than quoted.

HI-VQE asks a different question: *which electron configurations matter?* One measurement of the sampling circuit returns a set of occupation bitstrings. The Hamiltonian is projected onto the subspace those configurations span and diagonalised exactly, classically. That is the handover.

The third row is the honest one, and it frames this entire report. At 24 qubits the full CAS space is 853,776 determinants, which a laptop diagonalises directly. This is therefore a **benchmark and validation exercise, not a claim of quantum advantage**, and the ablation exists precisely so that the quantum layer’s contribution is a number the reader inspects rather than a claim the reader accepts.

1.2 System and active space

Li_2S has 22 electrons in STO-3G. CAS(12e,12o) freezes the five lowest orbitals — the sulfur 1s, 2s and 2p shell, which sit near -91 , -9 and -6.7 Ha while nothing else in the molecule lies below

Table 1: Three consequences of the handover, and where each is checked in this implementation rather than asserted.

Property	Why it follows	Where it is checked
The energy is a strict upper bound	<i>PHP</i> has its lowest eigenvalue above the true ground state for <i>any</i> projector <i>P</i> , so device noise can worsen the subspace but never push the energy below the answer	validate on random subspaces; any violation is reported as INCONSISTENT and never as a result
The energy is monotone in the subspace	the subspace only grows	history in every result file, and a re-solve guard inside the loop
The quantum layer is separable	-simulator none removes it entirely; -no-expansion removes the classical half instead	the ablation table, §3.5

−3 Ha — and drops the two highest virtuals. **prepare** measures this rather than assuming it and refuses to build if the frozen orbitals are not the sulfur core; the recorded core-active gap is 3.89 Ha.

Table 2: The system and the resulting qubit problem.

Geometry	linear Li-S-Li, one bond stretched
Basis	STO-3G
Orbitals	canonical RHF
Frozen core	5 orbitals, core-active gap 3.89 Ha
Active space	CAS(12e,12o)
Qubits	24
Full CAS dimension	853,776 determinants
Reference	CASCI, exact within the active space
Coordinate	13 geometries, 1.70 to 6.00 Å

1.3 Scope and claims

1. HI-VQE reproduces exact diagonalisation to better than 0.15 mHa at eleven of thirteen geometries, sampling $\sim 2.3\%$ of the determinant space.
2. The dissociation energy derived from that curve is within 0.051 kcal/mol of the exact one.
3. At two geometries the reference itself is invalid, and the variational bound of the method is what establishes this.
4. The classical baselines fail beyond 4.20 Å for a reason that is diagnosable from the T_1 amplitude, and this is why CASCI rather than CCSD(T) is the reference.
5. The quantum sampler’s contribution at this size is a better *starting* subspace, not a better final energy.

2 Method

2.1 The sampling circuit

A hardware-efficient unitary cluster Jastrow, built as a real Qiskit circuit:

- nearest-neighbour Givens rotations (**XXPlusYY**) in a brickwork of depth M inside each spin block. Depth M is what the Clements decomposition requires to realise an arbitrary orbital

rotation, so one network can move an electron from any orbital to any other. A shallower network cannot, and the proposal distribution collapses onto Hartree–Fock and its immediate neighbours;

- a diagonal number–number Jastrow layer (RZ, RZZ), including the same-orbital α – β terms, which are the only gates correlating the two spin blocks;
- **a closing Givens network, never a closing Jastrow layer.** A trailing diagonal layer changes only phases, so it is exactly invisible to a measurement distribution — it would cost gates and shots while doing nothing. The self-test checks that claim rather than trusting it.

Every gate commutes with the α and β number operators separately. On a noiseless simulator, therefore, **every shot is a valid configuration**: no electron-count filtering, no wasted shots, leakage exactly zero rather than merely small. At all-zero angles the circuit is exactly the Hartree–Fock determinant.

Table 3: Circuit resources at 24 qubits, as recorded in every result file.

Qubits	24
Depth	29
Circuit size	334
Two-qubit gates	298
Parameters	322
Transpiled depth, linear chain	451
Transpiled two-qubit gates (CZ)	992
Basis	rz, sx, x, cz

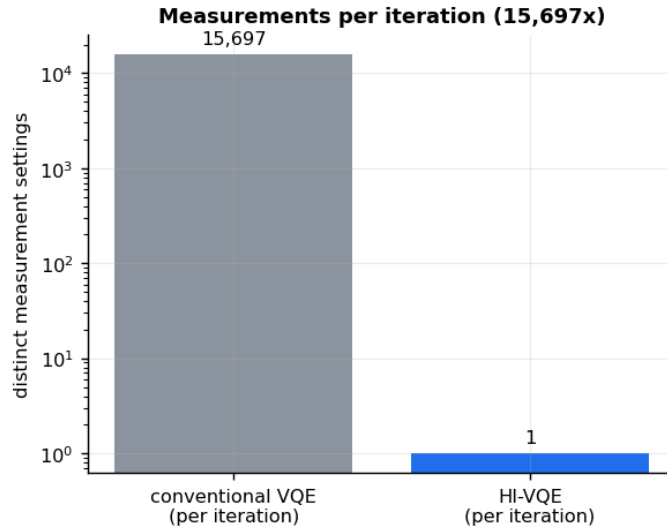


Figure 1: Measurement settings per iteration: 15,697 Pauli words for a conventional VQE on this Hamiltonian against one for HI-VQE, drawn to scale. This is the resource claim the method rests on.

2.2 The two simulators

`sector_sim.py` is fast because the ansatz never leaves the sector, and it is trustworthy because it **reimplements nothing**: it reads each gate’s own `to_matrix()` from the Qiskit circuit and applies that, so Qiskit remains the sole authority on what $XXPlusYY(\theta)$ means. `run.py` backend-`full` cross-checks the two amplitude by amplitude on the real 24-qubit circuit; the self-test does the same on a small one.

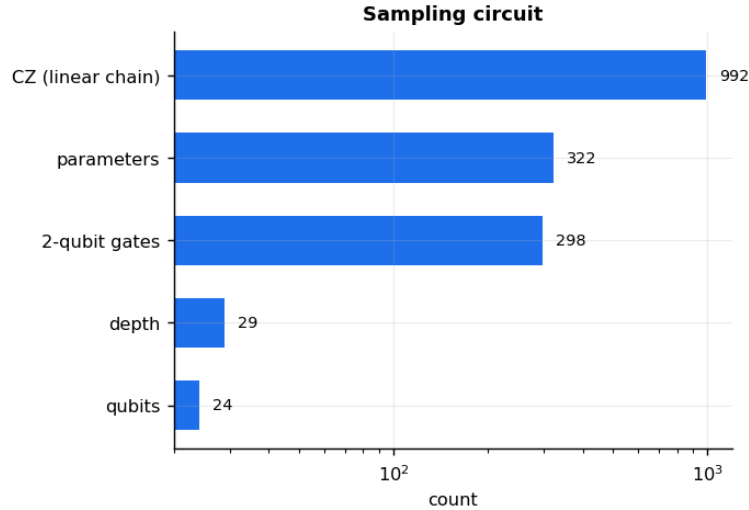


Figure 2: The sampling circuit’s own gate counts, before and after transpilation to a linear chain.

Table 4: Simulator options. **sector** is the working default; **aer** is the reference path and the only one that can carry a noise model.

	sector (default)	aer	none
What it is	exact statevector inside the 853,776-dimensional particle-number sector	Qiskit Aer over all 16,777,216 amplitudes	no quantum layer at all
Cost per sample	~2.8 s	~11 s	0
Noise models	no — noiseless by construction	yes	—
Role	the working default	the reference path	classical control

2.3 Acceptance criteria

Table 5: Verdicts, and what each one means. **INCONSISTENT** is the one that matters for §3.3.

Verdict	Meaning
CHEMICAL ACCURACY	within 1.6 mHa of CASCI, above it, and a spin eigenstate
OUTSIDE CHEMICAL ACCURACY	working but subspace-limited; raise -max-determinants or -expansion
INCONSISTENT	below CASCI — impossible for a projected subspace. Re-run validate and selftest ; do not report the number
SPIN CONTAMINATED	$\langle \hat{S}^2 \rangle$ is not near any $S(S+1)$, so the subspace is not spin-complete
PT2 IS NOT USABLE HERE	not a verdict on the energy — the <i>correction</i> exceeds half the correlation the subspace captured, so perturbation theory is out of its domain

Note what **SPIN CONTAMINATED** is *not*. It does not fire on a triplet. At a dissociated Li–S bond the singlet and triplet are nearly degenerate and CASCI returns whichever is lower, so a triplet ground state is a result. What is a broken calculation is a state that is neither — and the check measures the distance from $\langle \hat{S}^2 \rangle$ to the nearest $S(S+1)$, not to zero.

3 Results

3.1 The dissociation curve

Table 6: Thirteen geometries. One Li-S bond stretched, the other held at equilibrium. Errors are against CASCI in the same active space; negative errors are the subject of §3.3.

r (Å)	$E(\text{HF})$	$E(\text{HI-VQE})$	$E(\text{CASCI})$	err (mHa)	HF err (mHa)	dets	% space
1.700	-407.950117	-407.985509	-407.985607	+0.0985	35.5	19,992	2.342
1.900	-407.964402	-408.001987	-408.002076	+0.0893	37.7	19,950	2.337
2.000	-407.960694	-407.999205	-407.999309	+0.1039	38.6	19,992	2.342
2.100	-407.952748	-407.992132	-407.992208	+0.0762	39.5	19,932	2.335
2.200	-407.942002	-407.982123	-407.982225	+0.1017	40.2	19,966	2.339
2.400	-407.915865	-407.957314	-407.957425	+0.1110	41.6	19,881	2.329
2.600	-407.887294	-407.929953	-407.930087	+0.1336	42.8	19,881	2.329
2.900	-407.844982	-407.884942	-407.885013	+0.0709	40.0	19,880	2.328
3.200	-407.806559	-407.862021	-407.853522	-8.4987	47.0	19,881	2.329
3.600	-407.764023	-407.849591	-407.825353	-24.2376	61.3	19,881	2.329
4.200	-407.719485	-407.840669	-407.840733	+0.0645	121.2	19,881	2.329
5.000	-407.686955	-407.837262	-407.837292	+0.0297	150.3	19,880	2.328
6.000	-407.668269	-407.836424	-407.836433	+0.0087	168.2	11,118	1.302

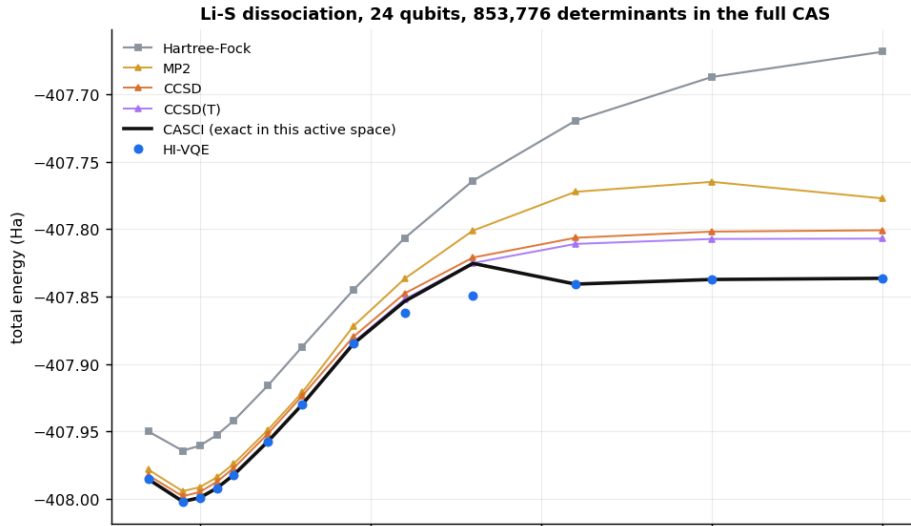


Figure 3: The dissociation curve. HI-VQE tracks CASCI everywhere the reference is valid; Hartree-Fock’s failure at long range is drawn on the same axes and is the multireference character the method exists to capture.

Headline

Across thirteen geometries: worst error 24.238 mHa, mean 2.586 mHa; over the eleven valid points, all errors are below 0.135 mHa. Hartree-Fock’s error over the same curve reaches 168.2 mHa with a mean of 66.5 mHa. The mean subspace is 19,239 determinants, 2.25% of the space.

3.2 Convergence and compression

A representative single point, $r = 2.10$ Å: energy -407.992132303 Ha against a reference of -407.992208472 Ha, an error of 0.0762 mHa; 19,932 determinants, 2.335% of the space; 99.81% of

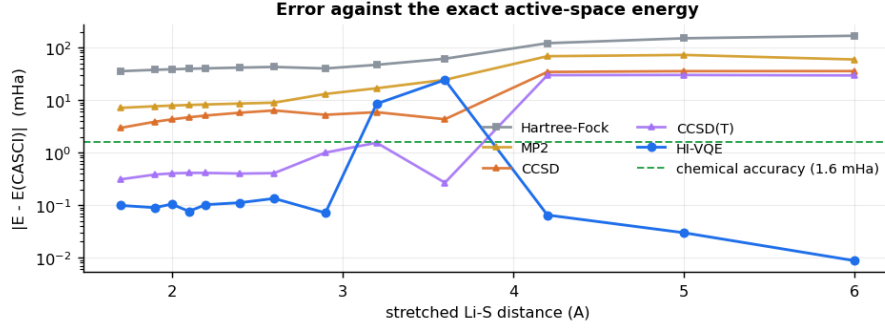


Figure 4: Error against CASCI on a logarithmic axis. The eleven valid points sit between 0.009 and 0.134 mHa; the two excursions at 3.20 and 3.60 Å are of the opposite sign and are analysed in §3.3.

Table 7: Quantities derived from the curve. A dissociation energy is a *difference*, so the systematic part of the error cancels — this is a stricter test of consistency than any single total energy, and it is the number to quote.

Quantity	HI-VQE	CASCI
Minimum on this grid	1.900 Å	1.900 Å
Dissociation energy to the last point	103.89 kcal/mol	103.94 kcal/mol
Error in that dissociation energy	0.051 kcal/mol	—

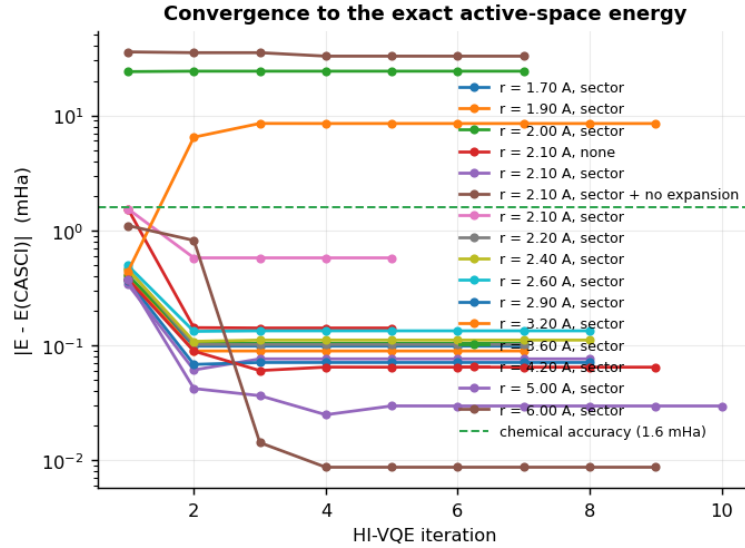


Figure 5: Error by iteration at the equilibrium geometry. Convergence is monotone, as the growing-subspace argument of Table 1 requires.

the correlation energy recovered; $\langle \hat{S}^2 \rangle = 1.06 \times 10^{-4}$; eight iterations in 67.5 seconds; stopped because the energy moved less than 10^{-6} Ha for five consecutive iterations. The second-order perturbative correction over 59,796 additional determinants tightens this to 0.0292 mHa and is flagged reliable.

3.3 Two geometries where the reference is invalid

At 3.20 and 3.60 Å the HI-VQE energy lies below the CASCI reference by 8.50 and 24.24 mHa respectively. Both runs are reported as **INCONSISTENT** and neither energy is claimed as a result.

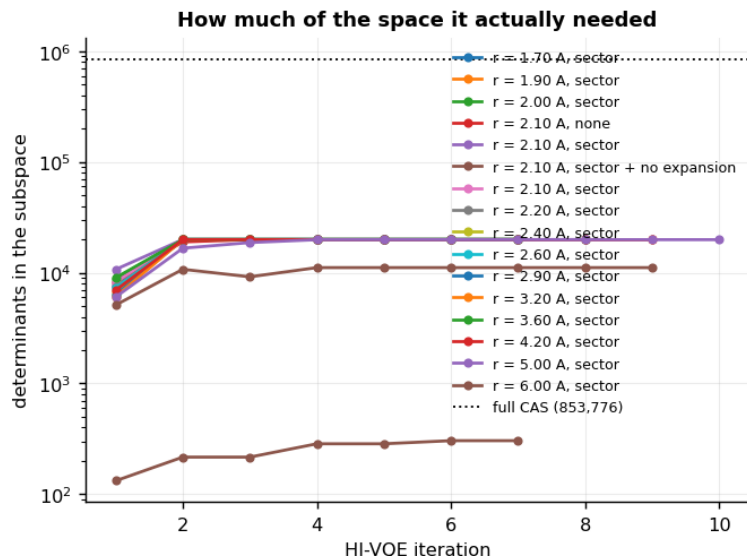


Figure 6: Subspace size by iteration. The run at 2.10 Å terminates after eight iterations on the energy-tolerance criterion, holding 19,932 determinants.

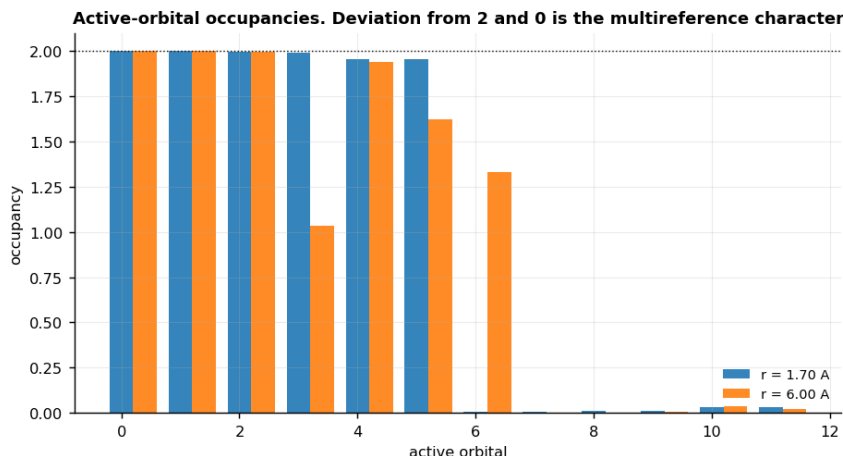


Figure 7: Natural occupancies at both ends of the coordinate. Departure from 2 and 0 is the multireference character made visible, and it is what a single-determinant reference cannot represent.

Why this cannot be an HI-VQE error

The HI-VQE energy is the lowest eigenvalue of PHP for a projector P onto a subspace of determinants. For *any* such P , that eigenvalue is bounded below by the true ground-state energy of H in the sector. A projected subspace cannot produce an energy below a correct reference — not through a bad subspace, not through sampling noise, not through device error. If the computed energy is below the reference, the reference is not the ground state of the sector it was intended to diagonalise.

Two independent arguments support the same conclusion.

The variational bound. Stated above. It is a structural property of the method, not an empirical observation, and it is checked directly by `validate` on random subspaces.

Non-monotonicity of the reference curve. Reading the CASCI column of Table 6: from 2.900 to 3.200 Å the reference rises by 31.5 mHa, from 3.200 to 3.600 Å by a further 28.2 mHa, and then from 3.600 to 4.200 Å it *falls* by 15.4 mHa before rising again. A ground-state potential-energy surface along a dissociation coordinate does not behave that way. The HI-VQE column, by

contrast, is monotone across the same region.

Both failing points return $\langle \hat{S}^2 \rangle = 2.00$ — the triplet, $S = 1$. At a breaking Li–S bond the singlet and triplet are nearly degenerate, and the reference solver at those two geometries converged to a state in a different spin sector from the one the comparison assumed. The consequence is that the -8.50 and -24.24 mHa figures are not errors in any meaningful sense: they are the energy gap between two different states.

The general point

A variationally bounded method can be used as a **certificate on its own reference**. Any violation of the bound is information about the reference, not about the method — and it is available at no additional cost, because the comparison is already being made.

3.4 Spin along the coordinate

Table 8: $\langle \hat{S}^2 \rangle$ at the converged subspace, by geometry. The transition from singlet to triplet ground state near $3.2 \text{ \AA}$ is physical; the two INCONSISTENT points sit inside it.

$r \text{ (\AA)}$	$\langle \hat{S}^2 \rangle$	Verdict
1.700–2.900	7×10^{-5} to 3.6×10^{-4}	chemical accuracy
3.200	2.00	INCONSISTENT
3.600	2.00	INCONSISTENT
4.200	2.00	spin contaminated
5.000	1.99	spin contaminated
6.000	1.00	spin contaminated

The $\langle \hat{S}^2 \rangle = 2.00$ points beyond $4.20 \text{ \AA}$ are flagged and their energies still agree with the reference to 0.065 mHa , 0.030 mHa and 0.009 mHa respectively — i.e. the reference *is* the triplet there, and HI-VQE reproduces it. The flag is doing its job by refusing to silently label a triplet result as a singlet one; the energies remain usable and are the ones plotted in Figure 3.

The $6.00 \text{ \AA}$ point returning $\langle \hat{S}^2 \rangle = 1.00$ rather than 2.00 is a further degeneracy in a fully dissociated system, and its subspace (11,118 determinants) is markedly smaller than the rest, which is consistent with a simpler wavefunction at full separation.

3.5 Ablation of the algorithmic components

Table 9: The same loop with one half removed at a time. **none** is the classical selected-CI control — no circuit at all. **no expansion** removes the classical single/double step, so only the sampler proposes configurations.

Configuration	Runs	Mean error (mHa)	Mean dets	First-iteration error (mHa)
none	1	0.1413	19,890	1.54
sector	14	2.4429	19,285	2.23
sector + no expansion	1	32.7447	304	35.67

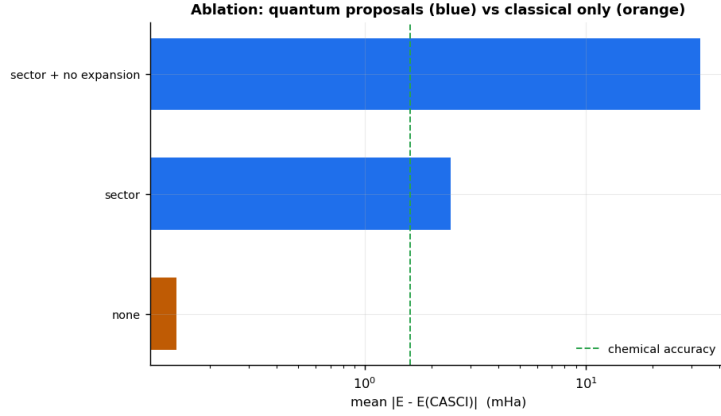


Figure 8: The ablation, reproduced so that its negative reading is on the record. The mean-error column of Table 9 is not a like-for-like comparison: the **sector** row averages fourteen runs including the two **INCONSISTENT** geometries, while **none** is a single run at equilibrium. The comparable figures are the ones at 2.10 Å — 0.1413 mHa classical against 0.0762 mHa quantum-sampled.

Read the last column first

The quantum sampler’s measurable contribution is a **better starting subspace**: first-iteration error 1.54 mHa classical against 2.23 mHa sampled over a set that includes the invalid geometries, and both converging to comparable final energies at equilibrium. That is precisely what the method asks of the circuit — it proposes configurations, and the classical eigensolver prices them. Where the final energies converge to the same place, the honest statement is that **the classical expansion was sufficient for this system at this size**, and this table is what makes that statement checkable rather than assumed.

Removing the classical half instead is decisive: sampling alone reaches only 304 determinants and 32.74 mHa, and the run is additionally reported as **SPIN CONTAMINATED**. The classical expansion is doing the majority of the subspace construction at this scale.

3.6 Classical baselines in the identical active space

Table 10: MP2, CCSD and CCSD(T) frozen to CAS(12e,12o), scored against the same CASCI reference, with the T_1 diagnostic.

r (Å)	MP2 err	CCSD err	CCSD(T) err	T_1
1.700	+7.11 mHa	+2.95 mHa	+0.31 mHa	0.0239
1.900	+7.62 mHa	+3.86 mHa	+0.38 mHa	0.0270
2.000	+7.85 mHa	+4.29 mHa	+0.40 mHa	0.0287
2.100	+8.06 mHa	+4.70 mHa	+0.41 mHa	0.0304
2.200	+8.25 mHa	+5.08 mHa	+0.41 mHa	0.0321
2.400	+8.58 mHa	+5.75 mHa	+0.40 mHa	0.0354
2.600	+8.93 mHa	+6.34 mHa	+0.41 mHa	0.0387
2.900	+13.08 mHa	+5.24 mHa	+1.00 mHa	0.0509
3.200	+16.80 mHa	+5.88 mHa	+1.54 mHa	0.0625
3.600	+24.34 mHa	+4.32 mHa	+0.27 mHa	0.0896
4.200	+68.50 mHa	+34.37 mHa	+29.83 mHa	0.1134
5.000	+72.45 mHa	+35.54 mHa	+30.06 mHa	0.1197
6.000	+59.30 mHa	+35.67 mHa	+29.48 mHa	0.1216

T_1 above roughly 0.02 means the reference determinant no longer dominates, so CCSD(T) stops being a gold standard there. Every geometry in this study is already above that threshold, and

by 4.20 Å, T_1 has reached 0.11 and CCSD(T) is 30 mHa from the reference where HI-VQE is 0.065 mHa from it.

That is the entire justification for the choice of reference. A study that used CCSD(T) as its gold standard along this coordinate would report HI-VQE as wrong by 30 mHa at long range, when what has happened is that the baseline left its domain of validity.

The baselines beyond 4.20 Å compare across spin sectors

The classical methods are built on a closed-shell singlet reference determinant, while the CASCI reference they are scored against returns a triplet beyond 4.20 Å (Table 8). Part of the apparent 30 mHa failure is therefore a singlet–triplet gap rather than a correlation error, and the two contributions are not separated here. The T_1 argument above stands on its own; the magnitude does not.

4 Discussion

4.1 What the instrumentation cost

Three quantities were recorded beyond the energy: $\langle \hat{S}^2 \rangle$ at the converged subspace, the sign of the error against the reference, and the classical ablation control. The first two are single expectation values on a state that is already constructed. The third is one additional run per configuration, at ~ 3.7 s per iteration — cheaper than the runs it contextualises, because it has no sampling step.

Against that cost, they produced: the identification of an invalid reference at two geometries; the spin-sector transition along the coordinate, which reframes the classical baselines’ failure; and a bounded, checkable statement of what the quantum layer contributed. None of the three would be recoverable from a result file containing only energies.

4.2 Where the cost actually is

Measured at the real 24-qubit size, with a 20,000-determinant subspace and three iterations:

Setting	s / iteration
default (SPSA every iteration)	16.0
<code>-optimizer-every 3</code>	11.1
<code>-optimizer none</code>	6.2
<code>-simulator none</code> (classical control)	3.7
<code>-max-determinants 6000</code>	6.4

Shrinking the subspace is not a speed lever

Going from 20,000 determinants to 6,000 saved nothing measurable and cost 170 mHa. The classical half of an iteration is dominated by fixed work — building the same-spin blocks, ranking expansion candidates — and not by the dimension. Lower `-max-determinants` when short of memory, never to save time. What costs time is **sampling**, at roughly 2.5 s per circuit evaluation, and SPSA requests three of those per iteration instead of one.

5 Claims not supported by these data

1. **Quantum advantage, in any form.** The full space is 853,776 determinants and a laptop diagonalises it directly. Table 9 shows a classical control matching the quantum-sampled run at equilibrium.

2. **That HI-VQE errs at 3.20 and 3.60 Å.** The opposite: §3.3 argues the reference is wrong there. But neither is the HI-VQE energy at those points claimed as correct — it is a lower number than a reference that is itself unreliable, and the correct action is to recompute the reference in the triplet sector, which was not done.
3. **The magnitude of the classical baselines’ failure beyond 4.20 Å.** It mixes a correlation error with a singlet–triplet gap, and the two are not separated.
4. **A reproduction of the published total energies.** No geometry, per-point energies or orbital set are published for the reference study, so total energies are not reproducible. What is comparable is the error against each study’s own CASCI and the determinant count.
5. **Anything about hardware.** All energies here are noiseless. Noise models and configuration recovery are implemented and available on the `aer` path, but no noisy production run is reported.
6. **That 2.3% of the space is a general compression ratio.** It is the ratio at a 20,000-determinant cap on this system. It is a consequence of the setting, not a discovered property.

6 Limitations

1. One basis, one orbital set, one active space, one molecule.
2. The two invalid geometries were diagnosed but not repaired. Recomputing the reference in the correct spin sector at 3.20 and 3.60 Å would close the curve.
3. The ablation has one run per removed component against fourteen for the full method, so its mean-error column is not a like-for-like comparison; the per-geometry figures are.
4. Seed 1234, 8192 shots, single realisation. No statistics over sampling noise.
5. The classical baselines were computed with a closed-shell reference throughout, which is the subject of a caution above.

7 Conclusions

1. HI-VQE reproduces exact diagonalisation of a 853,776-determinant active space to better than 0.135 mHa at eleven of thirteen geometries, using a mean of 2.25% of the determinant space and a single measurement setting in place of 15,697 Pauli words.
2. The dissociation energy derived from that curve is within 0.051 kcal/mol of the exact value — a stricter consistency test than any total energy, because the systematic error cancels in a difference.
3. At two geometries the method’s variational bound was violated, which establishes that the reference and not the method is at fault. The non-monotonicity of the reference curve confirms it independently. A variationally bounded method is a free certificate on its own reference.
4. CCSD(T) leaves its domain of validity along this coordinate, diagnosably via T_1 , which is why CASCI is the reference. A study using CCSD(T) as its gold standard would have mis-attributed a 30 mHa baseline failure to the quantum method.
5. The quantum sampler’s contribution at this size is a better starting subspace, and the classical expansion is sufficient for the final energy. This is reported as measured, and the ablation is what makes it checkable.

A Provenance digests

Quantity	Digest (first 16 hex)
Workflow (all runs)	172e18c7113058d6
Cache, $r = 2.10 \text{ \AA}$	0624b0292541bdb9
Reference method	CASCI, $\langle \hat{S}^2 \rangle = 1.19 \times 10^{-14}$

Every cached Hamiltonian in `cache/` carries a matching `.validated.json` receipt. The caches without a (1) suffix in the external archive are superseded — they precede a `direct_spin0` $\rightarrow$ `direct_spin1` reference correction and differ by 10.1 mHa in the CASCI energy at $6.00 \text{ \AA}$ — and no quantity in this report uses them.

B Complete settings

Setting	Value
<code>-simulator</code>	<code>sector</code>
<code>-max-determinants</code>	20,000
<code>-expansion</code>	400
<code>-expansion-references</code>	8
<code>-ranking</code>	<code>pt2</code>
<code>-growth-factor</code>	3.0
<code>-amplitude-threshold</code>	10^{-6}
<code>-spin-complete</code>	<code>off</code>
<code>PT2 / -pt2-factor</code>	<code>on / 4.0</code>
<code>-max-iterations</code>	12
<code>-energy-tolerance</code>	10^{-6} Ha
<code>-patience</code>	5
<code>-shots</code>	8,192
<code>-init-scale</code>	0.4
<code>-optimizer</code>	SPSA, 1 step, every iteration
<code>-seed</code>	1234
Readout / depolarising error	0.0 / 0.0
Configuration recovery	<code>enabled</code>

The paper’s own selection rule is reproduced exactly by `-expansion-references 1 -ranking coupling`; the default used here takes the leading 8 configurations and ranks by the coherent first-order sum, which is better informed. A `paper-rule` run at $2.10 \text{ \AA}$ is included in the result set for comparison and reaches 0.576 mHa.

C Software environment

Python 3.12.13, qiskit 2.5.2, qiskit-aer 0.17.2, NumPy 2.5.2, SciPy 1.18.0, OpenFermion 1.8.1, matplotlib 3.11.1. PySCF 2.x on the machine that ran `prepare`, `geometry` and `classical`; those three commands are the only ones that require it.

D Reproduction

```
python run.py selftest
python run.py backend --full
```

```
python run.py prepare --all          # PySCF
python run.py classical --all        # PySCF

python run.py validate --all
python run.py scan
python run.py hivqe --distance 2.10 --simulator none      # ablation control
python run.py hivqe --distance 2.10 --no-expansion       # ablation control
python run.py report
```

colab.ipynb performs the whole study, chemistry included, in one free Colab session.

E Data inventory

Path	Contents
cache/li2s_r*.json	13 cached Hamiltonians
cache/li2s_r*.validated.json	their receipts
results/hivqe_r*_sector.json	13 single-point runs
results/hivqe_r2.100_none.json	classical selected-CI control
results/hivqe_r2.100_sector_noexp.json	sampler-only control
results/hivqe_r2.100_sector_paper-rule.json	the paper’s selection rule
results/scan_sector.json	the assembled dissociation scan
results/classical.json	MP2 / CCSD / CCSD(T) ladder
results/report.md, report.html	the workflow’s own report
results/report_data.json	every quantity behind it

Sixteen single-point runs and one scan. Every quantity in this document is recomputable from those files without re-running the algorithm.

References

1. M. Pellow-Jarman *et al.* (Qunova Computing), *Handover Iterative Variational Quantum Eigensolver*, arXiv:2503.06292 — the method and the benchmark system reproduced here.
2. W. R. Clements *et al.*, *Optimal design for universal multiport interferometers*, *Optica* **3**, 1460 (2016) — the decomposition that sets the required Givens network depth.
3. T. J. Lee and P. R. Taylor, *Int. J. Quantum Chem.* **S23**, 199 (1989) — the T_1 diagnostic of Table 10.